

3.3 Derivatives Worksheets

****Remember that simplifying before attempting to find the derivative is worthwhile****

1. Using Leibniz notation $\left(i.e., \frac{dy}{dx}, \frac{dP}{dt}, etc.\right)$, find the derivatives for each of the following functions.

a) $y = x^8 - 3\sqrt{x} + \frac{5}{x^3}$

b) $q = 5x^e + \sin(e^\pi) - \ln(5) + 2e^x$

c) $y = e^x \left(\frac{1}{2}x^{10} - 2x \right)$

d) $w = x \left(\frac{x}{\sqrt{x^3}} \right)^4$

e) $y = \frac{2t}{5-t^5}$

f) $T = \frac{\sqrt[3]{k} + k^{-2}}{k}$

2. Suppose that $f(5)=1$, $f'(5)=6$, $g(5)=-3$, & $g'(5)=2$. Find each of the following:

a) $(fg)'(5)$

a) $\left(\frac{f}{g}\right)'(5)$

b) $\left(\frac{g}{f}\right)'(5)$

c) $\left(\frac{fg}{g-f}\right)'(5)$

3. What is the equation of line tangent to the graph of $y = 2\sqrt{x^3} + 3x^2 + x + 7$ at its y -intercept?
4. If $f(2) = 10$ and $f'(x) = x^2 f(x)$ for all x , find $f''(2)$.
5. Find the x -value for each point on the graph of $f(x) = \frac{5}{4}x^4 - \frac{14}{3}x^3 - 3x^2$ where the line tangent to the curve is parallel to the line $y = 0$.

6. Find the equations of the tangent lines to the curve $y = \frac{x^2}{x+1}$ that are parallel to the line $6x + 2y = 1$.

7. At what points, if any, does the graph of $y = x^3 - 3x + 4$ have a horizontal tangent?

8. Find equations of the tangent line and the normal line the curve $y = x\sqrt{x}$ at the point $(1,1)$.

9. Compute $f''(x)|_{x=1}$ if $f(x) = x^2(x-2)$.

10. Consider the function $y = 5x^{-1} - \frac{1}{5}x$.

a) Find the equation of the tangent line to the curve at the point $(5, 0)$.

b) Find the area of the triangle formed from the coordinate axes and the tangent line to the curve at the point.

TRUE/FALSE

1. If f and g are differentiable, then $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$.

2. If f and g are differentiable, then $\frac{d}{dx}[f(x)g(x)] = f'(x)g'(x)$

3. If $y = e^2$, then $y' = 2e$

4. The derivative of a polynomial is a polynomial.